## Objective Type Questions

11. The number which is subtracted from $\frac{-6}{11}$ to get $\frac{8}{9}$ is _____.

(A) $\frac{34}{99}$ (B) $\frac{-142}{99}$ (C) $\frac{142}{99}$ (D) $\frac{-34}{99}$

12. Which of the following pairs is equivalent?

(A) $\frac{-20}{12}, \frac{5}{3}$ (B) $\frac{16}{-30}, \frac{-8}{15}$ (C) $\frac{-18}{36}, \frac{-20}{44}$ (D) $\frac{7}{-5}, \frac{-5}{7}$

13. $\frac{-5}{4}$ is a rational number which lies between _____ .

(A) 0 and $\frac{-5}{4}$ (B) –1 and 0 (C) –1 and –2 (D) –4 and –5

14. Which of the following rational numbers is the greatest?

(A) $\frac{-17}{24}$ (B) $\frac{-13}{16}$ (C) $\frac{7}{-8}$ (D) $\frac{-31}{32}$

15. The sum of the digits of the denominator in the simplest form of $\frac{112}{528}$ is _____ .

(A) 4 (B) 5 (C) 6 (D) 7

## 1.3 Basic Arithmetic Operations on Rational Numbers

All the rules and principles that govern fractions in the basic operations apply to rational numbers also.

### 1.3.1 Addition

There can be four different situations while doing addition.

**Type 1 : Adding numbers that have same denominators**

This is simply like adding like fractions and the result is the sum of the numerators divided by their common denominator.

**Example 1.10**

Add : $\frac{-6}{11}, \frac{8}{11}, \frac{-12}{11}$

***Solution:***

Write the given rational numbers in the standard form and then add them.

So, $\frac{-6}{11} + \frac{8}{11} + \frac{-12}{11} = \frac{-6+8-12}{11} = \frac{-10}{11}$

**Type 2 : Adding numbers that have different denominators**

After writing the given rational numbers in the standard form, use the LCM of their denominators to convert the numbers into equivalent rational numbers with a common denominator so that this reduces to Type1.



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